

1. DEFINITION

DEFINITION
Lauroyl macrogol-6 glycerides EP / Lauroyl polyoxyl-6 glycerides NF / Lauroyl macrogolglycerides (6) ChP

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Lauroyl Polyoxylglycerides Hydrogenated palm/palm kernel oil PEG-6 esters
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	8EPU9MJ01K (HYDROGENATED PALM/PALM KERNEL OIL PEG-6 ESTERS)
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE
This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (oral, topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Alcoholysis reaction between PEG-6 and Hydrogenated Palm/Palm kernel oils

COMPOSITION	
Well defined multi-constituent substance composed of mono-, di- and triglycerides and PEG-6 mono- and di- esters of Lauric acid (C12)	
Full Labeling (Key letters system) $A_1 \geq 75.0$; $50.0 \leq A_2 < 75.0$; $25.0 \leq B < 50.0$; $10.0 \leq C < 25.0$; $5.0 \leq D < 10.0$; $1.0 \leq E < 5.0$; $0.1 \leq F < 1.0$; $G < 0.1$	
Lauroyl Polyoxylglycerides Hydrogenated palm/palm kernel oil PEG-6 esters	$A_1 \geq 75.0\%$
Other constituents (i.e. incidental ingredients)	
None	



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4. DRUG MASTER FILE

Country	Master file number
US DMF (Type IV – Excipient)	5182
For submission of the DMF in another country, please contact your local Gattefossé representative	

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	Conforms to monograph
European Pharmacopeia Current edition	Lauroyl macrogol-6 glycerides
USP/ NF Current edition	Lauroyl polyoxyl-6 glycerides
Chinese Pharmacopoeia Current edition	Lauroyl macrogolglycerides (6)

6. SPECIFIC PHARMACEUTICAL REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	Ingredient Name	Use	Restrictions
Approved	Lauroyl macrogolglycerides Lauroyl polyoxylglycerides	Available for use as an Excipient Ingredient in: Biologicals, Export Only, Prescription Medicines	/
AUSTRALIA: Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homoeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	NHPID name	Route of administration/ Restrictions
Approved Chemical Name	Lauroyl polyoxylglycerides	Role Non-medicinal Purposes: Solubilizing Agent
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		



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7. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSÉ					
Study type/ duration	Route	Species	Test article	Results/Conclusions	Reference
Acute irritation (Patch test)	Dermal	Human	LABRAFIL® M2130CS 30% in stone oil	Non Irritant Cutaneous irritation index: 0.00	(Gattefossé data, GAT-9117, 1991)
Acute irritation (JORF)	Ocular	Rabbit	LABRAFIL® M2130CS 20% in water	Non Irritant Ocular irritation index: 0.00	(Gattefossé data, GAT-7516, 1975)
Sensitisation (Marzulli and Maibach's)	Dermal	Human	LABRAFIL® M2130CS 20% in stone oil	Not Irritant Not sensitizing	(Gattefossé data, GAT-98393, 1998)
Acute toxicity	Oral	Rat	LABRAFIL® M2130CS pure	LD50 _(oral) > 2000 mg/kg	(Gattefossé data, GAT-8815, 1988)
Ames test (OECD 471)	In vitro	S. typhimurium	LABRAFIL® M2130CS	Negative. Not mutagenic.	(Gattefossé data, GAT-98398, 1998)

CIR COMPENDIUM (Cosmetic Ingredient Review) (http://www.cir-safety.org)			
Denomination	Reference	Maximum "as used" concentration	Conclusions
Hydrogenated Palm/Palm Kernel Oil PEG-6 Esters	IJT 33 (Suppl 4):13-39, 2014	Up to 24 %	The CIR expert panel concluded the Hydrogenated Palm/Palm Kernel Oil PEG-6 Esters is safe in the present practices of use and concentration in cosmetics when formulated to be nonirritating.

8. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
INCI NAME	CAS #	EINECS #	Chemical name
Lauroyl Polyoxylglycerides	57107-95-6 (or 9004-81-3 + 27638-00-2)	Not applicable (or Not applicable + 248-586-9)	Poly(oxy-1,2-ethanediyl), α,α',α'' -1,2,3-propanetriyltris(ω - hydroxy-, dodecanoate (Or Poly(oxy-1,2-ethanediyl), α -(1-oxododecyl)- ω -hydroxy- + Dodecanoic diester with 1,2,3-propanetriol:



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NATIONAL INVENTORIES					
INCI NAME	U.S.A. (TSCA)	Canada (DSL/NDSL)	Australia (AIIC)	China (IECSC)	Inventory names
Lauroyl Polyoxylglycerides	Listed	NDSL listed	Listed	Listed	Poly(oxy-1,2-ethanediyl), α,α',α''-1,2,3-propanetriyltris(ω-hydroxy-, dodecanoate: IECSC, NDSL, REACH, TCSI, TSCA, VNECI, AIIC (Or) Poly(oxy-1,2-ethanediyl), α-(1-oxododecyl)- ω-hydroxy-: AIIC, DSL, ENCS, NZIoC, PICCS, REACH, TCSI, TSCA, VNECI, IECSC, AREC, ECL + Dodecanoic diester with 1,2,3-propanetriol: AIIC, DSL, ENCS, IECSC, NZIoC, PICCS, TCSI, TSCA, EINECS, AREC, ECL, VNECI
U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics... are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting. Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation. Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities. Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration. Chemicals not listed on the AIIC must be assessed before importation. Australia: AIIC - Canada: DLS/NDSL - China: IECSC - European Union: EINECS, REACH - Japan: ENCS - Mexico: INSQ - New Zealand: NZIoC - Philippines: PICCS - South Korea: ECL, AREC - Switzerland: SWISS - Taiwan: TCSI - Thailand: TDCA - USA: TSCA - Vietnam: VNECI					

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
INCI NAME	Status of the product	Status per substance
Lauroyl Polyoxylglycerides	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACH regulation.

9. HISTORY OF USE

9.1. Pharmaceutical applications

9.1.1. FDA IIG reference for Chemical equivalent

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
HYDROGENATED PALM/PALM KERNEL OIL PEG-6 ESTERS* (UNII: 8EPU9MJ01K)	TOPICAL	Emulsion, Cream	5 %	NA
LAUROYL POLYOXYLGLYCERIDES	ORAL	Capsule	218 mg	NA



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FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
(UNII: NA)	ORAL	Tablet	0.15 mg	
	ORAL	Tablet, film coated	0.15 mg	
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

**removed from IIG in April 2019 due to new data standard. The reference was not removed from IIG due to safety concerns.*

9.1.2. FDA IIG references to the most similar chemical (read-across approach)

In the **read-across approach**, endpoint information for one chemical (the source chemical) is used to predict the same endpoint for another chemical (the target chemical), which is considered to be "similar" in some way (usually on the basis of structural similarity or on the basis of the same mode or mechanisms of action). In principle, read-across approach can be used to assess physicochemical properties, toxicity, environmental fate and ecotoxicity.

For any of these endpoints, read-across approach may be performed in a qualitative or quantitative manner.'

The similarities may be based on the following:

- a common functional group (e.g. aldehyde, epoxide, ester, specific metal ion);
- common constituents or chemical classes, similar carbon range numbers;
- an incremental and constant change across the category (e.g. a chain-length category);
- the likelihood of common precursors and/or breakdown products, via physical or biological processes, which result in structurally similar chemicals.

These similarities justify data gap filling in a chemical category by applying read-across.

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
LAUROYL PEG-32 GLYCERIDES (UNII: H5ZC52369M)	ORAL	CAPSULE		7200mg
	ORAL	TABLET	0.15mg	
STEAROYL POLYOXYL-32 GLYCERIDES (UNII: G6EP177239)	ORAL	CAPSULE		960 mg
	ORAL	CAPSULE, EXTENDED RELEASE		8 mg
	ORAL	TABLET		3 mg
STEAROYL POLYOXYLGLYCERIDES (UNII: NA)	ORAL	CAPSULE	260 mg	
	ORAL	TABLET	23.34 mg	
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				



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9.1.3. FDA IIG references for larger family of chemicals

The following reference covers multiconstituent substances containing Lauroyl macrogol-6 glycerides.

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
PEG VEGETABLE OIL* (UNII: Not available)	IM – SC	Injection	7 %	NA
	IV(INFUSION)	Injection	5 %	NA
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

*removed from IIG in July 2019 due to new data standard. The reference was not removed from IIG due to safety concerns.

9.1.1. Other references

HANDBOOK OF PHARMACEUTICAL EXCIPIENT (Current edition by R. C. Rowe, P.J. Sheskey, W. G. Cook, M. E Fenton)			
Denomination	Functional Category	Regulatory status	Safety
Polyoxylglycerides	Dissolution enhancer; emulsifying agent; nonionic surfactant; penetration agent; solubilizing agent; sustained release agent.	Lauroyl polyoxylglycerides and stearoyl polyoxylglycerides are approved as food additives in the USA. Included in the FDA Inactive Ingredients Database (oral route: capsules, tablets, solutions; topical route: emulsions, creams, lotions; vaginal route: emulsions, creams). Oleyl polyoxylglycerides are included in a topical cream formulation licensed in the UK.	Polyoxylglycerides are used in oral and topical pharmaceutical formulations, and also in cosmetics and food.

Liability clause

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